

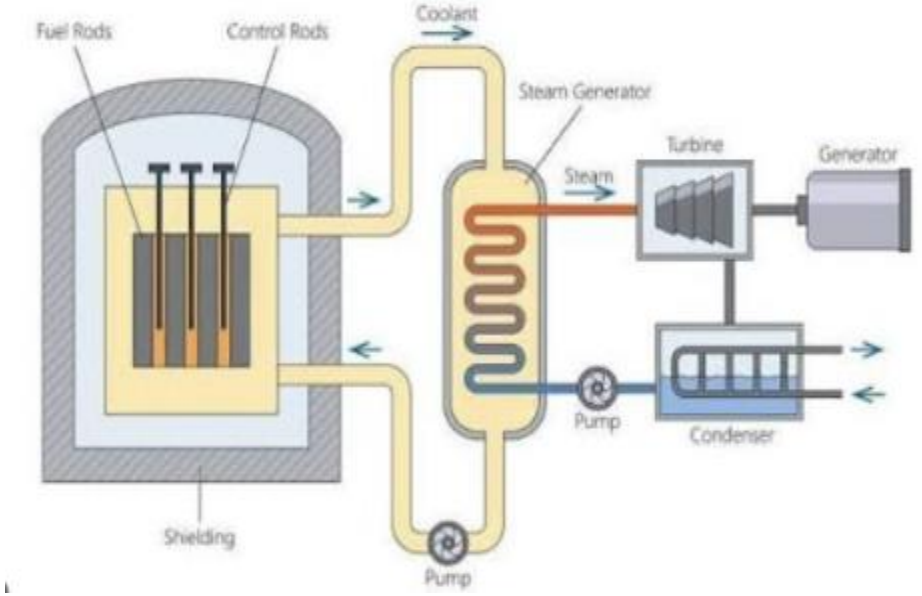
Comparative Study of PWR, BWR and PHWR Nuclear Reactors

(Working Principle, Layout, Technical Comparison, Safety and Applications)

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Basic Working of a Nuclear Power Plant

- A nuclear power plant is essentially a steam power plant in which the heat required to produce steam is obtained from nuclear fission instead of burning coal.
- Nuclear Fission → Heat Generation → Steam Production → Turbine → Generator → Electricity
- Nuclear fission: Fuel such as Uranium-235 undergoes controlled fission inside the reactor core and releases a large amount of heat.
- Heat transfer: A coolant carries heat away from the reactor core. Depending on reactor type, water may directly boil or transfer heat through a steam generator.
- Steam generation: The heat converts water into high-pressure steam.
- Power generation: Steam expands through a steam turbine, which rotates an electrical generator.
- Condensation & recirculation: Exhaust steam is condensed in a condenser, and the water is pumped back into the cycle.



Main Parts Used in PWR, BWR and PHWR

Reactor core
- Contains the **nuclear fuel** where fission takes place.

Nuclear fuel
- Provides fissile material for energy generation.
- Typical examples:
- PWR → Enriched uranium
- BWR → Enriched uranium
- PHWR → Usually natural uranium

Moderator
- Slows down fast neutrons so that the chain reaction can continue efficiently.
- PWR/BWR → Light water
- PHWR → Heavy water, D_2O

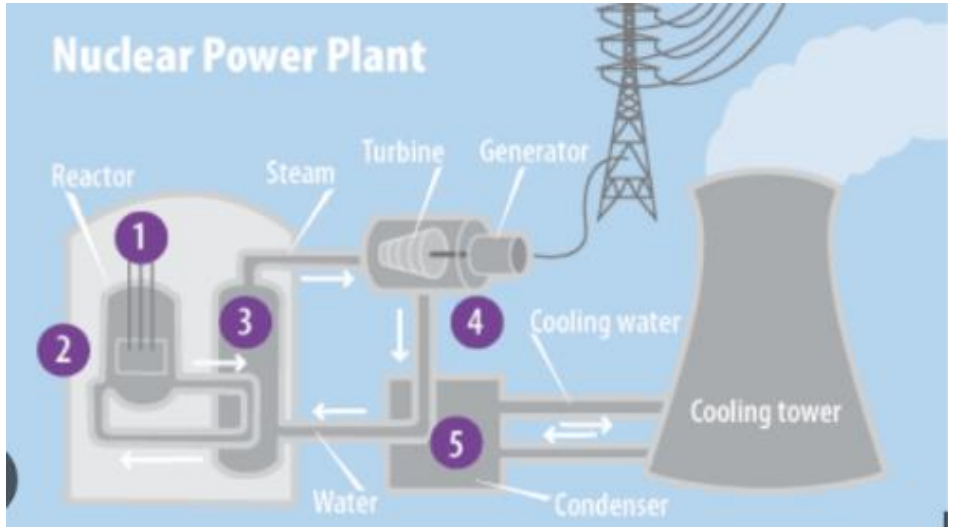
Coolant
- Removes heat generated in the reactor core.
- PWR → Pressurized light water
- BWR → Light water
- PHWR → Pressurized heavy water

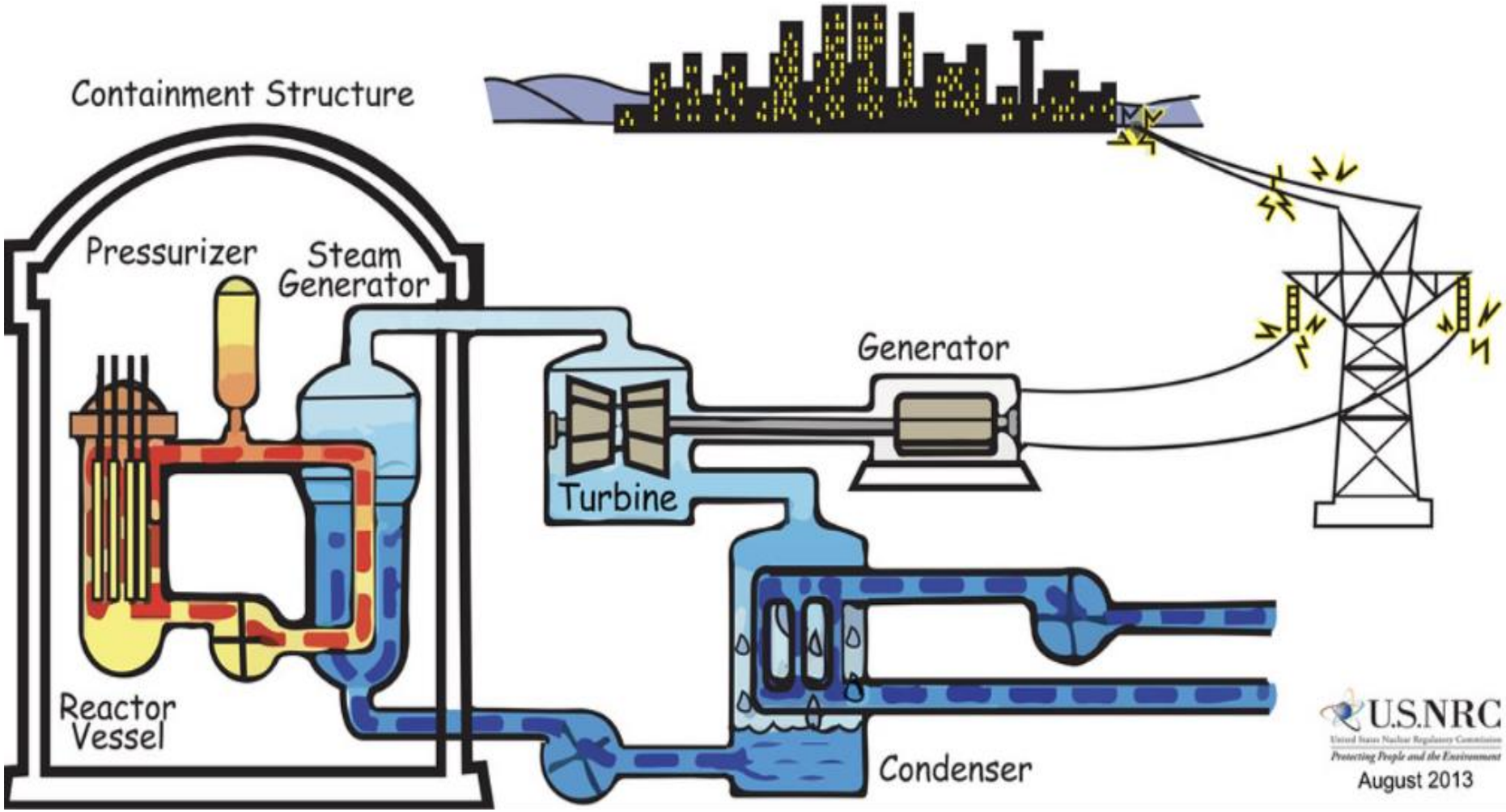
Steam turbine
- Converts steam energy into mechanical energy.

Generator
- Converts mechanical energy into electrical energy.

Condenser
- Condenses turbine exhaust steam into water.

Cooling system
- Rejects waste heat to atmosphere/water source.





Pressurized Water Reactor (PWR): Layout

Working of PWR

Principle
In a PWR, water in the primary circuit is maintained at **very high pressure**, preventing it from boiling inside the reactor.

Process

- Nuclear fission heats the reactor coolant.
- High-pressure water carries heat from the reactor to the **steam generator**.
- Primary water does **not boil**.
- Heat passes to secondary-loop water.
- Secondary water converts into steam.
- Steam drives the turbine-generator.
- Steam is condensed and returned to the steam generator.

Key characteristics

Moderator: Light water

Coolant: Light water

Fuel: Enriched uranium

Steam generation: Separate steam generator

Number of main fluid circuits: Two

PWR

Advantages

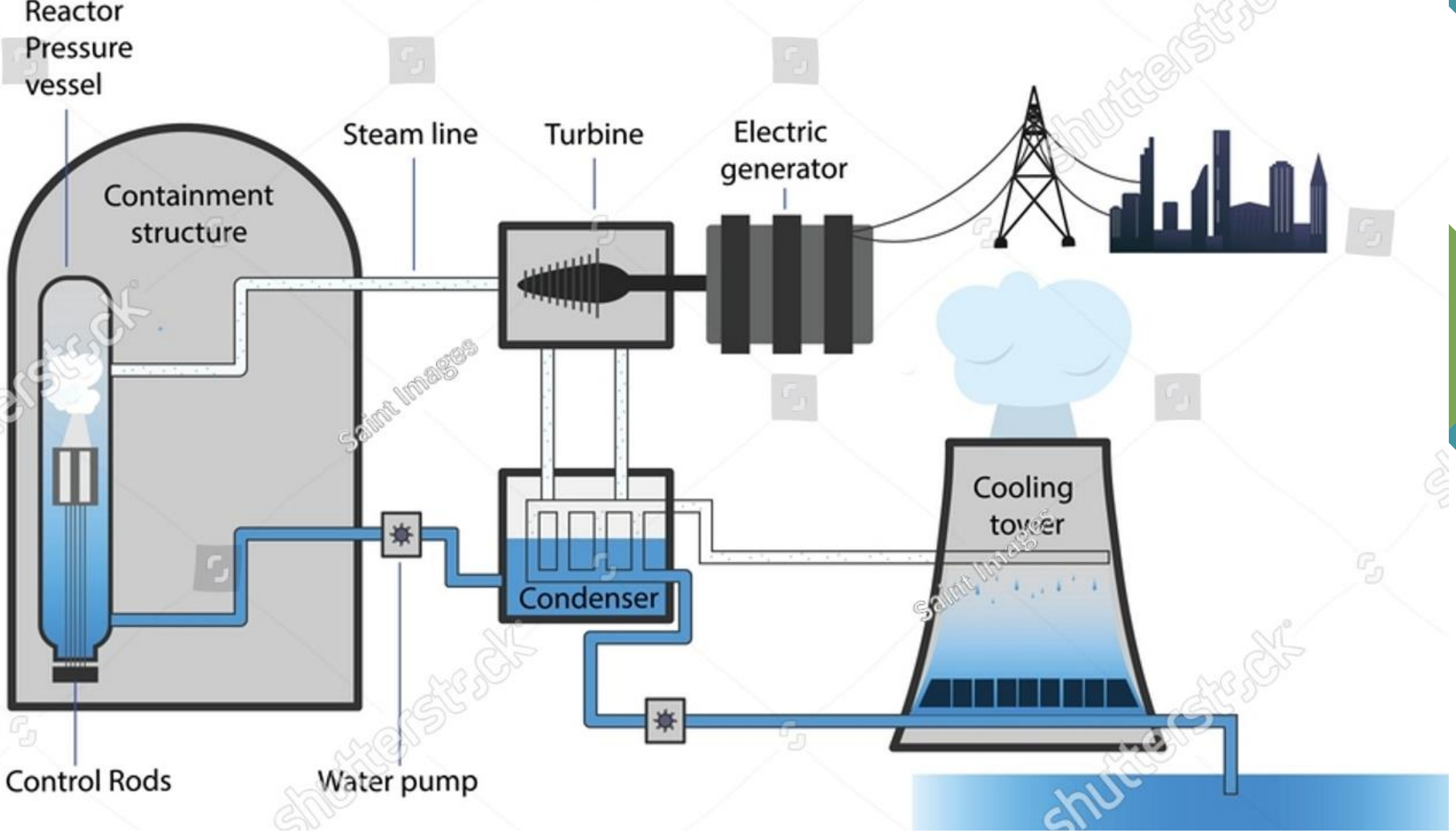
- Well-established technology
- Separate primary and secondary circuits
- Good control of reactor conditions

Limitations

- High-pressure reactor system Requires enriched fuel
- Complex steam generator and pressure system

Applications / Suitability :

- Large-scale commercial nuclear power generation
- Countries with established enriched-fuel infrastructure
- Continuous base-load electricity generation



Boiling Water Reactor (BWR)

Working of BWR

- Nuclear fission occurs in the reactor core.
- Water acts as both **coolant and moderator**.
- Heat from the fuel causes water to **boil inside the reactor vessel**.
- Steam rises through the reactor vessel.
- Steam separators and dryers remove water droplets.
- Dry steam flows directly to the **turbine**.
- Turbine drives the generator to produce electricity.
- Exhaust steam is condensed and pumped back to the reactor.

Key Feature:

- No separate steam generator is required.

Main advantage:

- Maintain good posture

Main concern:

- The turbine-side system can be exposed to radioactive contamination because reactor steam reaches the turbine.

BWR

Advantages

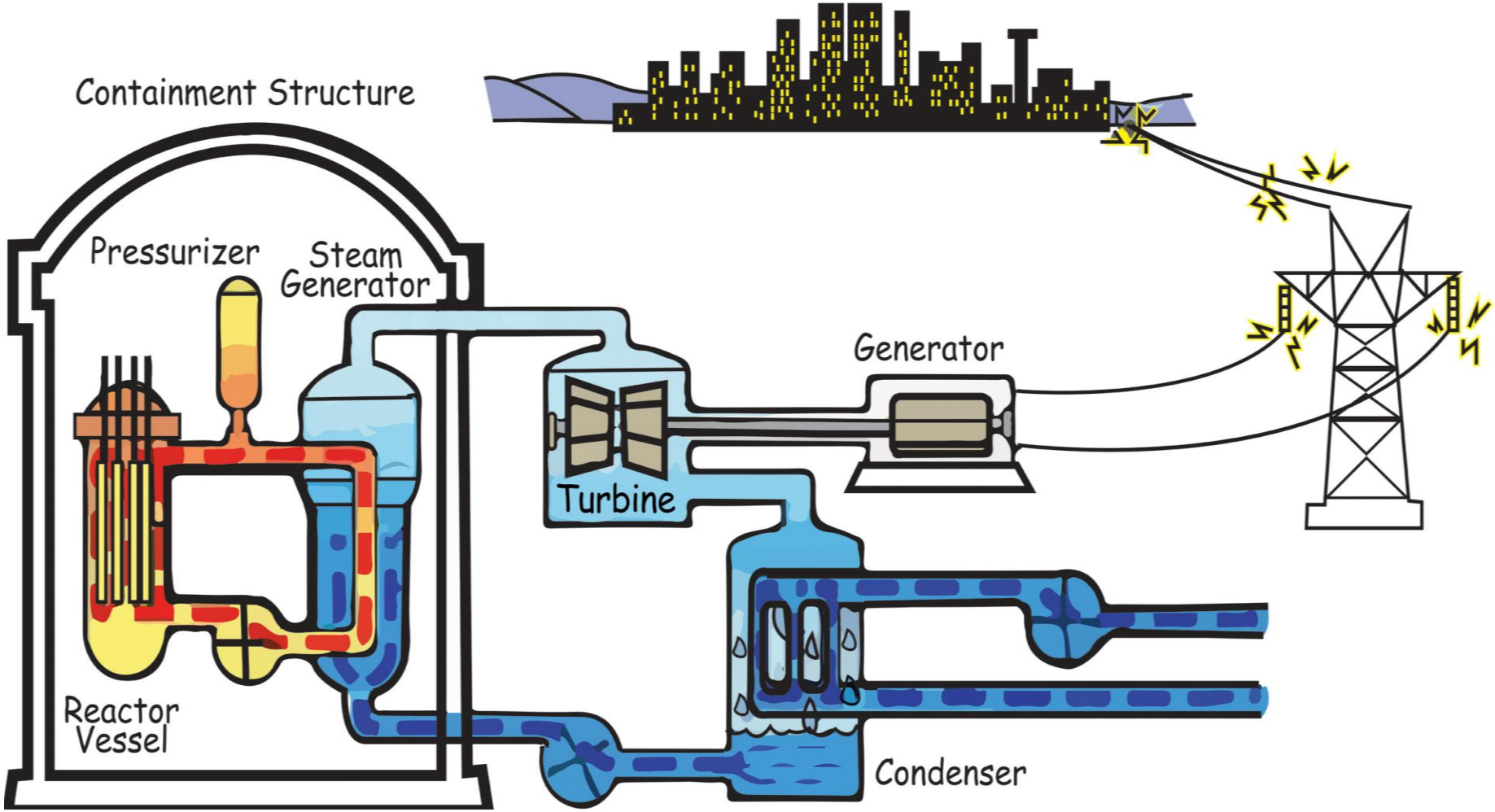
- Direct steam generation
- No separate steam generator
- Simpler primary steam cycle

Limitations

- Radioactive steam reaches turbine system
- More complex radiation protection around turbine
- Reactor control and water chemistry require careful management

Applications / Suitability :

- Large-scale electricity generation
- Plants where direct steam-cycle operation is preferred
- Utility-scale nuclear power generation



Pressurized Heavy Water Reactor (PHWR)

Working of PHWR

Nuclear fission takes place inside fuel bundles located in **pressure tubes**.

- **Heavy water (D_2O)** acts as moderator and coolant.
- Pressurized heavy-water coolant flows through the pressure tubes.
- Heat generated by fission is transferred to the coolant.
- Hot heavy water flows to the **steam generator**.
- Heat is transferred to ordinary/light water on the secondary side.
- Steam produced in the steam generator drives the turbine.
- The turbine drives the generator to produce electricity.
- Steam is condensed and returned to the steam generator.

Key Feature:

- PHWR can use **natural uranium fuel**, reducing the need for uranium enrichment.

PHWR

Advantages

- Can use natural uranium
- Heavy water provides good neutron economy
- Pressure-tube design allows online refuelling

Limitations

- Heavy water is expensive
- Large number of pressure tubes and associated components
- More complex maintenance of pressure-tube systems

Applications / Suitability :

- Countries having limited uranium-enrichment infrastructure
- Efficient utilization of natural uranium
- Large-scale base-load power generation

PWR vs BWR vs PHWR

Parameter	PWR	BWR	PHWR
Coolant	Light water	Light water	Heavy water
Moderator	Light water	Light water	Heavy water
Steam generation	Indirect	Direct	Indirect
Steam generator	Required	Not required	Required
Reactor pressure	High	Lower than PWR	High in coolant channels
Fuel	Enriched uranium	Enriched uranium	Natural uranium possible
Pressure vessel	Large reactor vessel	Large reactor vessel	Pressure tubes
Turbine steam	Non-radioactive secondary steam	Reactor steam	Non-radioactive secondary steam
Refuelling	Usually shutdown	Usually shutdown	Can be done online in many designs

Safety Features in These Reactors

Common Safety Features:

- **Control rods:** Absorb neutrons and control reactor power.
- **Emergency shutdown system:** Rapidly stops the nuclear chain reaction.
- **Emergency core cooling:** Provides cooling during loss-of-coolant conditions.
- **Containment structure:** Helps prevent release of radioactive material.
- **Multiple cooling systems:** Maintain safe core temperature.
- **Radiation monitoring:** Detects abnormal radiation levels.
- **Backup power systems:** Maintain essential safety equipment during power loss.

Conclusion

- **PWR, BWR and PHWR** all generate electricity using controlled nuclear fission.
- The major difference is the **coolant, moderator and method of steam generation**.
- **PWR:** Water remains pressurized and transfers heat to a separate steam circuit.
- **BWR:** Water boils directly inside the reactor and produces steam for the turbine.
- **PHWR:** Heavy water is used as coolant and moderator, and heat is transferred through steam generators.
- PWR and BWR mainly use enriched uranium, while PHWR can operate with **natural uranium**.
- The choice of reactor depends on **fuel availability, technology, safety requirements, cost and national energy strategy**.

Final takeaway:

PWR = Indirect steam generation
BWR = Direct steam generation
PHWR = Heavy-water reactor + pressure tubes

Thank you

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